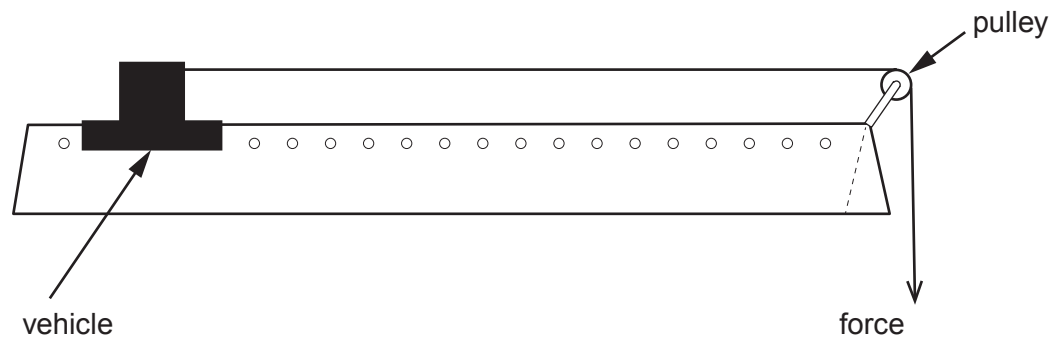


4. In a class experiment, a group of students use a force to accelerate a vehicle along a frictionless air track.



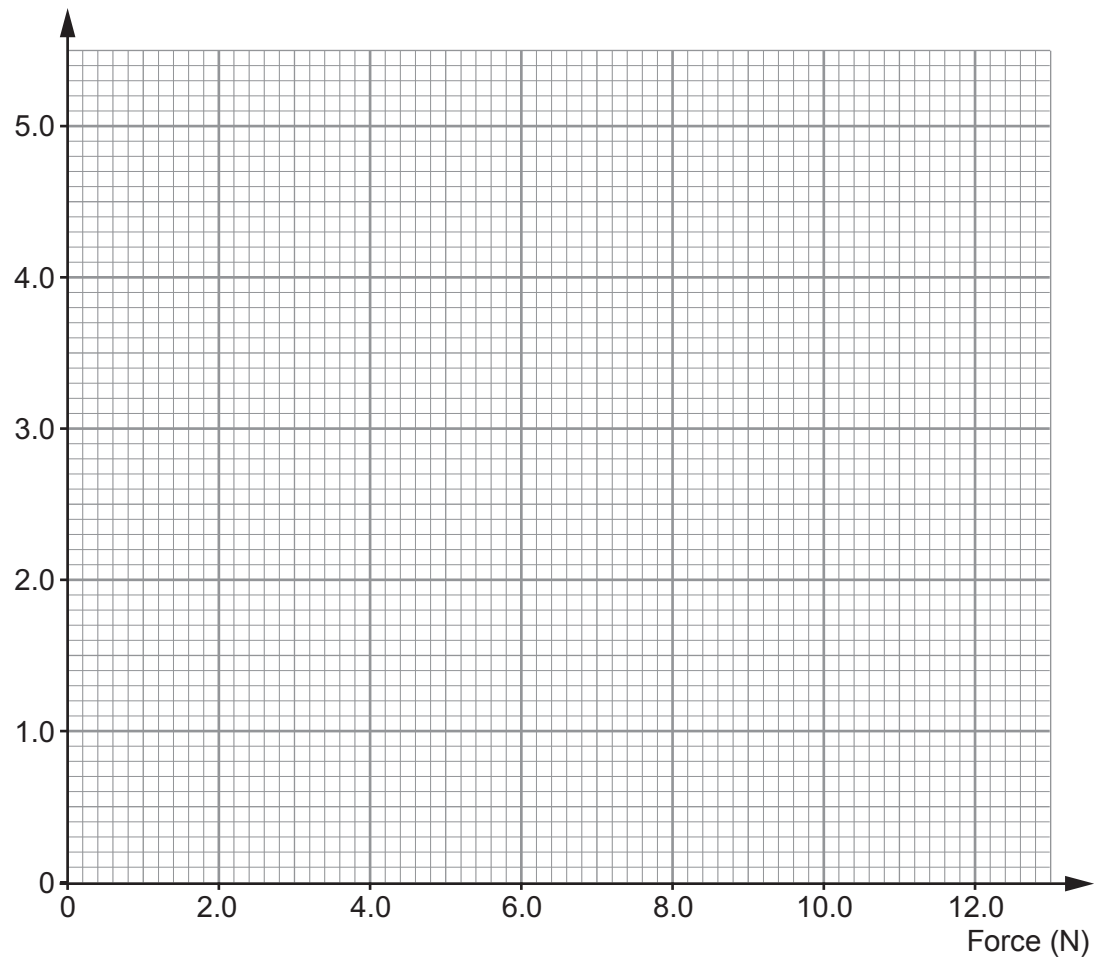
The acceleration of the vehicle is measured by a computer and the results are shown below.

Force (N)	2.0	4.0	8.0	10.0	12.0
Acceleration (m/s^2)	0.8	2.6	3.2	4.0	4.8

- (a) Plot the data on the grid below and draw a suitable line.

[3]

Acceleration (m/s^2)



- (b) (i) Use the graph to find the force that produces an acceleration of 2.0 m/s^2 . [1]

Force = N

- (ii) Use the equation:

$$\text{mass} = \frac{\text{resultant force}}{\text{acceleration}}$$

to calculate the mass of the vehicle. [2]

Mass = kg

- (iii) Another group of students in the class carry out the same experiment but they use a vehicle of greater mass. **Draw a line on the graph** to show how the acceleration would change for the same values of force. [1]

- (c) Eric suggests that if a force of 16 N is applied to the original vehicle the acceleration will be 5.6 m/s^2 . Use the data to consider whether this suggestion is correct. [2]

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